

- 1 (a) The gas molecules exert no intermolecular forces on one another except during collisions.

(b) (i) 1.  $p_B V_B = n_B R T_B$

$$\begin{aligned} n_B &= \frac{p_B V_B}{R T_B} \\ &= \frac{(2.0 \times 10^5)(3.0 \times 10^{-2})}{(8.31)(300)} \\ &= 2.4067 = 2.4 \text{ mol (shown)} \end{aligned}$$

2. Solution 1

$$\begin{aligned} \text{total internal energy} &= U_A + U_B \\ &= \frac{3}{2} p_A V_A + \frac{3}{2} p_B V_B \\ &= \frac{3}{2} (3.0 \times 10^5)(2.0 \times 10^{-2}) + \frac{3}{2} (2.0 \times 10^5)(3.0 \times 10^{-2}) \\ &= 18000 \text{ J} \end{aligned}$$

Solution 2

$$\begin{aligned} \text{total internal energy} &= U_A + U_B \\ &= \frac{3}{2} n_A R T_A + \frac{3}{2} n_B R T_B \\ &= \frac{3}{2} (1.8)(8.31)(400) + \frac{3}{2} (2.4)(8.31)(300) \\ &= 17949.6 = 17900 \text{ J} \end{aligned}$$

- (ii) 1. Total internal energy is conserved as this container is a closed system.

$$\text{total internal energy} = U_A' + U_B' = 17900$$

$$\frac{3}{2} n_A R T + \frac{3}{2} n_B R T = 17900$$

$$\frac{3}{2} R T (n_A + n_B) = 17900$$

$$T = (17900) \left( \frac{2}{3} \right) \left( \frac{1}{(8.31)(1.8 + 2.4)} \right)$$

$$= 341.91 = 342 \text{ K}$$

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2. As gas A is at a higher pressure than gas B, it expands and does work against the pressure of gas B, hence work done  $W$  on gas A is negative.

As gas A is at a higher temperature than gas B, heat is transferred from gas A to gas B, hence heat supplied  $Q$  to gas A is negative.

Since  $\Delta U = Q + W$ , the increase in internal energy  $\Delta U$  is negative which implies that the internal energy of gas A decreases. Since  $\Delta U \propto \Delta T$  (or

$U = \frac{3}{2}nRT$ ), the temperature of gas A decreases.

- (iii) Vacuum does not contain any particles and has no pressure. Hence in expanding, gas A does no work and does not transfer heat to any other body. Hence by the first law of thermodynamics, there is no change in internal energy and no change in temperature.